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SOLVAY - BRUSSELS
DICHLOROETHANE CHLORINATION
CHLORINATION IN THE LIQUID PHASE

THE MANUFACTURE OF 1-2 DICHLOROETHANE BY THE CHLORINATION
OF ETHYLENE IN THE LIQUID PHASE

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BRUSSELS
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ETHYLENE IN CONTACT WITH A BOILING LIQUID PHASE

SUMMARY

Different aspects of the chlorination of ethylene into 1-2 dichloroethane (EDC) in contact with a liquid phase are reviewed. In this note results are also given of experimental studies carried out in 1964 and continued in 1968 ("boiling chlorination").

A very rapid reaction takes place between the dissolved ethylene and chlorine. Its rate is determined by the rate of transfer of ethylene at the gas-liquid interface.

The main side reaction which gives rise to the formation of 1,1,2-trichloroethane (β -tri) can be easily controlled by the addition of ferric chloride and the introduction of oxygen into the reaction medium.

When the reaction is carried out in contact with a liquid phase maintained at its boiling point, the heat liberated by the reaction makes it possible to vaporise the EDC made and thus to obtain it free from ferric chloride and from β -tri. This makes it possible to avoid caustic washing, azeotropic drying and purification from heavy by-products necessary in the classic process. Furthermore since the quantity of heat liberated by the reaction is considerably greater than that necessary to vaporise the EDC made the excess can be used to purify the crude EDC from oxychlorination and/or the EDC

recycled from cracking. A substantial saving in steam can thus be achieved. However the impurities present in the EDC recycled from cracking react partially with chlorine. The yield with respect to chlorine is thus lowered by 3% which from the point of view of economics is equivalent to approximately 60% of the heat which can be recovered.

In principle introduction of crude EDC from oxychlorination on its own into the reactor should not lower the chlorine yield. In this case one can recover 55% of the heat liberated by (direct) chlorination with a reflux ratio of 0.77.

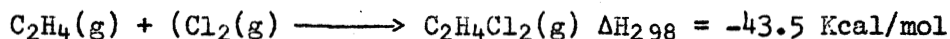
1 INTRODUCTION

A Current Processes

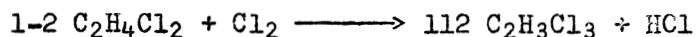
The manufacture of vinyl chloride from ethylene and chlorine is carried out in two stages;

- preparation of 1,2-dichloroethane (EDC)
- cracking of EDC

The chlorination of ethylene to EDC is an exothermic reaction which at 25°C liberates 43.5 Kcals/mole of EDC manufactured (Notice DGT/RECH 63.298 Note FN 1094 reference 1).



The addition chlorination of ethylene is accompanied by substitution chlorination which converts part of the EDC into β -tri.



The substitution yield is defined as being the molar relation

$$S = \frac{\beta\text{-tri}}{\text{EDC} + \beta\text{-tri}} \times 100$$

The substitution yield can be calculated from the HCl produced; in this way a substitution yield from HCl is obtained which in the majority of cases is close to that based on the amount of β -tri formed.

In the conventional process the reaction takes place in the presence of an inventory of EDC not at its boiling point, the heat liberated being removed by indirect cooling. The EDC produced contains amongst other things chlorine hydrogen chloride and ferric chloride; it is washed with an aqueous solution of caustic soda, then dried by azeotropic distillation. Crude EDC is thus obtained. Before being cracked crude EDC is distilled to eliminate heavies.

In the boiling process the reaction also takes place in an inventory of liquid EDC; the heat liberated is used to vaporise the EDC made. This way of operating makes it possible to eliminate the caustic wash, azeotropic drying and the elimination of heavies, which were indispensable in the conventional process. Furthermore since the quantity of heat liberated by the reaction is considerably greater than that necessary to vaporise the EDC made the excess can be used to purify the crude EDC from oxychlorination and/or to purify the EDC recycled from cracking. A considerable saving in steam can thus be achieved. Nevertheless boiling chlorination is of interest only if its yield is not appreciably less than that of the conventional process.

Finally for the sake of completeness we should mention that gas phase chlorination has been considered from time to time but does not seem to have been applied on an industrial scale.

B Conventional Process

Little information exists on the techniques and apparatus used industrially for the chlorination of ethylene in a liquid phase. However three variations can be made out.

- 1 Dissolving chlorine in liquid EDC followed by bubbling ethylene through the solution.
- 2 Stage wise introduction of chlorine and ethylene in circulating EDC (4), (5) (Notice DMP761 of 30.6.1966).
- 3 Simultaneous introduction of chlorine and ethylene into liquid EDC (6) (7). Most often ethylene is introduced in slight excess to the chlorine.

The Bayer process uses two reactors (see the scheme given below):* chlorine is introduced into the main reactor; ~~ethylene is introduced into the main reactor~~; ethylene is introduced into the secondary reactor in slight excess compared with the chlorine. EDC carrying dissolved chlorine passes from the main to the secondary reactor where it is dechlorinated. The reaction thus takes place in the presence of an excess of chlorine in the main reactor (achieving the chlorination of the ethylene) and in the presence of an excess of ethylene in the secondary reactor (exhaustion of the chlorine).

The reactors used have various configurations:

- Reactor empty or filled with packing.
- Reactor with or without an agitator.
- Cooling by a jacket by a coil or by circulation through an external cooler.

* Fig 1 At the back of the Translation

	Literature	Tavaux
Yield based on ethylene:	94 to 98.5%	97.5 to 98.5%
Yield based on chlorine:	97 to 100%	99 to 100%
8 ^{EDC} tri content of crude EDC product:	95 to 99.5%	99.5 to 99.8%

C Boiling Process

Recent patents often concern operating conditions which enable the EDC to be recovered in the form of vapour free from ferric chloride (5) (9) (11) (12) (13) (14). The EDC obtained thus is already partly purified from 'heavies'.

Union Carbide and Huls propose that the reactor should be surmounted by a rectifying column which makes a better separation of heavies possible. Union Carbide mentions that it is necessary in order to achieve a good reaction, to renew the EDC inventory in the reactor every six months.

Ethyl installed a boiling reactor in Greece which was started up at the beginning of 1967.

D Gas Phase Process

Several patents describe the chlorination of ethylene in the gaseous phase but it seems that there has been no industrial application. This process does not seem to be of much interest especially because the ethylene used is extremely pure (9). Experimental work was carried out by SBA in a pilot plant for several months (10). The reaction is normally catalysed by metallic salts deposited on a support (chlorides of Ca, Ba, Fe, Pb, deposited on pumice, asbestos activated carbon ...).

E Contents of the Note

In the present note different aspects of the manufacture of EDC by the chlorination of ethylene in contact with a liquid phase are examined.

In particular the results of an experimental study of boiling chlorination (carried out in 1968) and the industrial experience of Ethyl in this field are presented (Chapter III).

Chapter II is concerned with the conventional process (used at Tavaux and Jemeppe).

II THE CHLORINATION OF ETHYLENE IN LIQUID KEPT BELOW ITS BOILING POINT

A RATE, MECHANISM AND ORDER OF REACTION

In spite of the considerable industrial exploitation of the manufacture of 1,2-dichloroethane by the chlorination of ethylene in liquid EDC the mechanism of the reaction has not yet been completely explained.

Up to temperatures of 150°C chlorine and ethylene do not react together in a homogeneous gas phase (15). The reaction, however, takes place in contact with vessel walls and in contact with suitable catalysts. The absence of a gas phase reaction is confirmed by the simultaneous presence of chlorine and ethylene in the residual gases from chlorination. This was observed in the new direct chlorinator installed by Ethyl at Houston. (See Chapter II D).

In the presence of a suitable liquid medium the reaction is rapid at ambient temperature; it takes place between dissolved ethylene and dissolved chlorine. Doraiswamy (16) showed that in a reactor with an agitator, the rate of reaction is controlled by the rate of transfer of ethylene at the gas liquid interface; he found that

$$R = 1.92 \times 10^{-5} V_{gA}^{2/3}$$

Where R is moles of C_2H_4 reacting/hr cm^2 ,

V_{gA} cm/hr upward velocity of C_2H_4 brought to the free section of the reactor.

The mass transfer coefficient k_A^1 is related to the upward velocity of the ethylene by the relation

$$k_A^1 = 0.16 V_{gA}^{2/3}$$

This is in good agreement with the results obtained by introducing the reactants through nozzles into reactors without agitators (17 and 18).

According to Doraiswamy

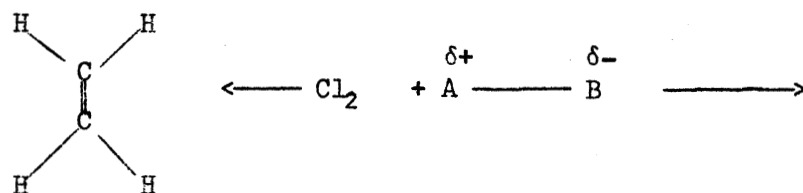
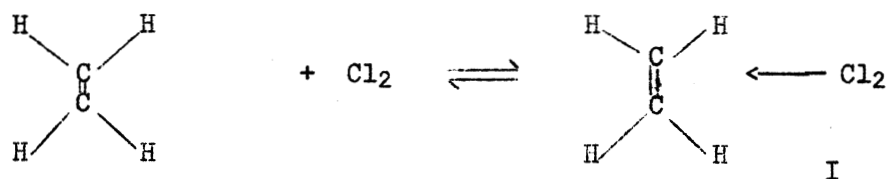
- The mass transfer coefficient of the ethylene is not affected by the presence of bubbles of chlorine.
- For the conditions studied the rate of reaction increases with the rate of agitation until a maximum value is obtained above which the rate of agitation has no further influence.
- The rate of reaction is independent of temperature between 33 and 56°C.

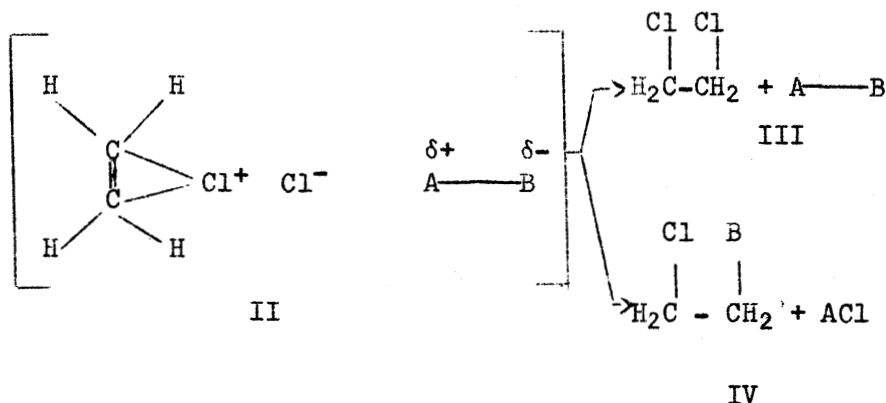
The addition of chlorine on to ethylene in a homogeneous liquid phase probably takes place according to a polar mechanism.

At ambient temperature shaded from light the rate of addition of chlorine on to ethylene or a parent olefin is practically zero in carbon tetrachloride, benzene and cyclohexane (19) (20). Increasing the polarity of the medium by the addition of substances such as hydrochloric acid or carboxylic acid appreciably increases the rate of reaction (19). In practice the presence of compounds liable to give off protons seems to be much more effective than increasing the polarity of the medium. In the case of carbon tetrachloride the acceleration of the reaction which does not start until after an induction period of from 1 to 1½ hours is attributed to the formation of a chlorination product (which is polar). It is however more probable that it is the hydrochloric acid (hydrogen chloride?) liberated by a secondary reaction which initiates the main reaction.

According to the laboratory studies carried out by Ethyl in a glass reactor containing EDC the reaction does not take place in the absence of iron (21). Norrish and Jones (22) shewed that the presence of liquid or absorbed water on the walls of a glass reactor accelerated the reaction between gaseous chlorine and ethylene. The same thing happens when the vessel walls were covered with cetyl alcohol or stearic acid. On the other hand in the presence of a covering of paraffin the reaction is very slow even when the water is introduced in the form of vapour (22) (23).

Due to interference from secondary reactions it has not been possible to determine the order of the reaction with certainty. Nevertheless it seems that the following reaction mechanism can be proposed:





I is a charge transfer complex

II is a transition complex (eventually a pair of ions stabilized by the group A—B).

In the case of the chlorination of ethylene A B can be for example: HCl, FeCl₃, H⁺FeCl₄⁻ the solvent (one or several molecules), etc.... For the general case A B can be a catalyst of the Lewis acid type such as FeCl₃, AlCl₃, SbCl₃ etc.....

When A B represents a non-ionisable group only the addition complex III can be formed. The reaction rate in a homogenous solution can therefore be written

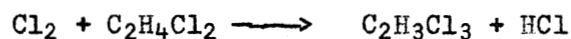
$$V = K[\text{C}_2\text{H}_4][\text{Cl}_2][\text{A B}]$$

The reaction order is therefore 1 with respect to C₂H₄, Cl₂, and AB.

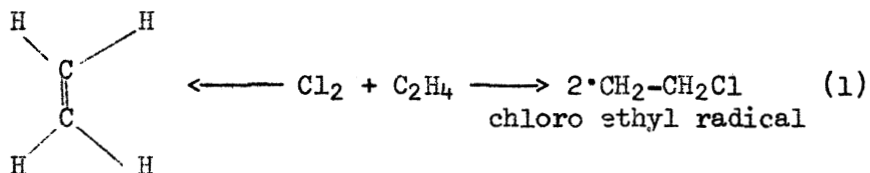
The radical addition of chlorine to ethylene which gives EDC takes place in parallel to the polar addition. However when the degree of substitution is small (see I-A) the contribution of this mechanism to the formation of EDC must be small. On the other hand when the degree of substitution is high an appreciable quantity of the EDC can be formed by the radical addition of chlorine to ethylene (9) (24).

B FORMATION OF 1,1,2-TRICHLOROETHANE

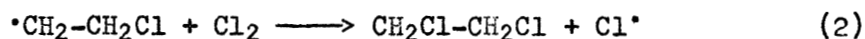
1,1,2-trichloroethane (β-tri) is the second most important product resulting from the chlorination of ethylene in 1,2-dichloroethane; it is formed from chlorine and EDC:



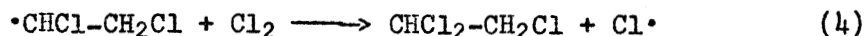
It is a radical, substitution reaction the initiation of which probably results from the action of ethylene on the charge transfer complex (24) already mentioned in paragraph II A:



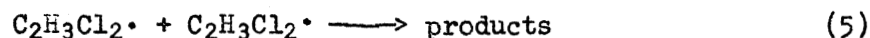
The activation energy of this reaction will be of the order of 6 kcal/mol (25). The chloroethyl radical reacts with a chlorine molecule:



The chain propagates according to the classic mechanism

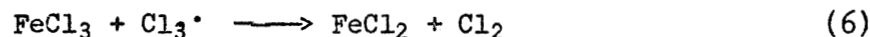


At ordinary temperatures in the absence of any foreign substance the chain is terminated by (Note FN 1184)

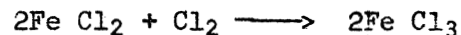


In practice it is not this reaction which limits the formation of β -tri: this is done by inhibitors such as oxygen and ferric chloride present in the reaction medium.

In the case of ferric chloride, the chain is ruptured according to the following mechanism (Note PBn^o 1151)



It should be noted that in an aqueous medium the FeCl_2 is rapidly reconverted into FeCl_3 by reacting with chlorine



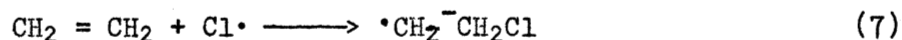
The initiation of radical chlorination by ethylene has also been demonstrated by Stewart and Hanson who obtained hexachlorobenzene by bubbling ethylene through a solution of chlorine in benzene (23).

REMARK 1

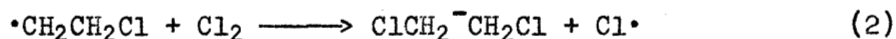
At normal temperatures in the absence of light the radical initiation $\text{Cl}_2 \longrightarrow 2 \text{Cl}^\bullet$ does not occur. In practice a solution of chlorine in EDC can be kept for several days without the chlorination starting.

REMARK 2

The radical reaction



which gives rise to EDC according to



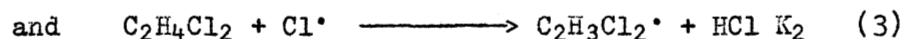
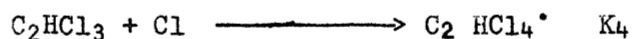
is much more rapid than reaction (3) consequently even in the absence of inhibitors the formation of β -tri should be very small. Nevertheless the ethylene concentration is almost zero in the liquid phase (the ethylene being consumed by the main reaction) and it is reaction (3) which does this. The effectiveness of oxygen as an inhibitor inspite of its slight solubility arises from the fact that it is uniformly distributed throughout the liquid phase and that it is not eliminated by the main reaction.

The presence of ethylene derivatives such as tri and perchloroethylene must limit the formation of β -tri by terminating the radical chains. However in the presence of ferric chloride the inhibiting effect of these ethylene derivatives is marked.

REMARK 3

To eliminate trichloroethylene present in EDC from cracking ICI ANZ introduced EDC containing TCE into their plant direct chlorinator. Under these conditions 60% of the trichloroethylene reacted (26).

The relative reactivity of trichloroethylene and EDC with respect to $\text{Cl}^\bullet$ according to the reaction



has been determined (Note PB 1151). The value found for the ratio K_4/K_2 is approximately 120.

Let S, %, be the degree of substitution

r, %, be the degree of conversion of C_2HCl_3 into C_2HCl_5

one has

$$\frac{100-r}{100} = \left(\frac{100-s}{100} \right)^{120}$$

From this it follows that for the 60% of the trichloroethylene to react the degree of substitution in the direct chlorinator must be about 0.8%. In industrial direct chlorinators the normal degree of substitution is 0.3%; then only 30% of the tri-chloroethylene is converted into pentachloroethane. On the other hand to eliminate 90% of the trichloroethylene a degree of substitution of 1.9% in the direct chlorinator would be necessary.

C THE EFFECT OF OXYGEN AND OF FERRIC CHLORIDE ON THE DEGREE OF SUBSTITUTION

When the chlorination of ethylene takes place in a glass reactor in the absence of oxygen and of ferric chloride the amount of β -tri formed is very high. The presence of ferric chloride and/or oxygen in the reaction medium limits considerably the formation of β -tri as is shown by the results given below (Notice PCR 65.218)

EDC circulation flow : 100 ltrs/hr
 C_2H_4 flow : 4.2 mol/hr
 Cl_2 flow : 4 mol/hr
 Temperature : 45°C

Reactor made from Pyrex Glass protected from the action of light (Note FN 1184)

Run No	Fe Cl_3 * added mg/ltr	O ₂ % vol C_2H_4	Degree of Substitution
1	Traces	0	36.5
2	100	0	4.2
3	250	0	1.89
4	Traces	2	2.7
5	100	2	0.08
6	250	2	0.08
* FeCl ₃ added to the circulation and not FeCl ₃ actually dissolved			

It is therefore beneficial to add FeCl_3 and oxygen simultaneously to limit the formation of β -tri. Normally oxygen is introduced with the reactants in the form of air; pure oxygen is slightly more effective.

Trials carried out on the pilot plant at Jemeppe in the presence of ferric chloride but with oxygen free chlorine (vaporised chlorine) confirmed the laboratory results obtained under the same conditions. The degree of substitution is particularly high when chlorine free from oxygen is introduced into the EDC circulation before the ethylene (Jemeppe notes No 341 and No437).

Tests carried out at the Shri Ram Institute of Bombay in a pilot plant in the absence of FeCl_3 (oxygen content unknown) gave a degree of substitution of 7.7%. The addition of ferric chloride made it possible to lower the degree of substitution to 1 to 2% (30).

Under industrial conditions (Ethyl - ICI and Solvay) only the addition of air has been carried out. At Tavaux the quantity of oxygen present in the chlorine used is sufficient (0.3 to 1.1% vol); no addition of air is necessary. For its part Ethyl add air to the extent of 2-5% of the volume of the chlorine when they use vaporised (oxygen free) chlorine (Note DT 7413 and Notice DG MP No 295). Under industrial conditions the introduction of ferric chloride is not necessary; dosage due to corrosion in the reactor is sufficient (see IIG) to maintain a ferric chloride concentration of 30-60 mg/kg in the circulation. The ferric chloride is formed in situ by the combined action of chlorine, air and moisture on the mild steel walls of the reactor. The same observation was made in the Jemeppe pilot plant. (Jemeppe Note 437). These quantities of oxygen and of ferric chloride clearly less than those mentioned in the laboratory tests enable the degree of substitution to be lowered to 0.2 to 0.5%.

Ethyl feel that the presence of metallic iron (wall) is more important than the presence of ferric chloride but that nevertheless the simultaneous presence of iron and ferric chloride is more effective than iron or ferric chloride on its own. Ethyl think that the iron could either favour the dissolution of FeCl_3 or promote the formation of a complex between ferric chloride and EDC (31).

More simply the metal wall could remove the water which could complex the ferric chloride, a complex which would not contribute to the inhibition of the radical reaction. A trial carried out in a glass reactor shewed that when water is introduced continuously into the reaction medium containing ferric chloride the degree of substitution approached the value obtained with oxygen on its own (Note FN 1184). When chlorinating ethylene with dilute chlorine in a scrubber irrigated with EDC there was a marked lowering in the degree of substitution when the reactants and the apparatus

were rigorously dried.

During a boiling chlorination trial carried out by Ethyl in laboratory apparatus water was introduced accidentally and the degree of substitution climbed to 8%. The ferric chloride probably passed into the aqueous phase and no longer contributed to the inhibition of the radical reactions. In the presence of oxygen the effect of water on the formation of β -tri is relatively slight; it can explain in certain cases an increase in the degree of substitution from 0.2-0.5% to 0.8-1.5%. Trials carried out at Jemeppe shewed that it was preferable to use anhydrous ferric chloride rather than $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ "which had the disadvantage of bringing with it water a substitution catalyst". (E.A.T. Jemeppe May 1963).

Hydrated ferric chloride is not necessarily inactive to $\text{Cl}\cdot$ radicals in chlorination. In practice FeCl_3 and $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ are equally effective as agents for the termination of chains in the polymerisations of acrylonitrile, methacrylonitrile and styrene in non-aqueous mediums (34). However, in the presence of water (and perhaps HCl and Cl_2) it can form droplets rich in ferric chloride. The ferric chloride thus extracted from the organic phase no longer contributes to the inhibition of radical reactions.

When the ferric chloride content is high (more than 0.5 gm/kg) the effect of oxygen becomes negligible and it is therefore no longer necessary to introduce it with the reactants (E.A.T. of Jemeppe June 1963).

The favourable effect of large quantities of ferric chloride on the lessening of the substitution to form β -tri was also shewn during chlorination under boiling conditions (see III-I).

As already mentioned above in conventional industrial chlorination the quantity of ferric chloride produced by corrosion is sufficient to limit substitution. The introduction of additional quantities of ferric chloride necessitates the installation of equipment for the introduction and metering of FeCl_3 ; furthermore there is the risk of causing difficulties in phase separation during the caustic washing of the EDC product.

Since the role of ferric chloride is essentially in its inhibiting action with respect to radical reactions it is useless to increase the ferric chloride content with a view to accelerating the main reaction. The addition of extra quantities of FeCl_3 to the reaction medium can only thus be envisaged when the degree of substitution is abnormally high.

D THE EFFECT OF THE MOLE RATIO $C_2H_4 : Cl_2$ ON THE DEGREE OF SUBSTITUTION

In general in industrial practice the ethylene is introduced in slight excess to the chlorine. This method of operating makes it possible to consume almost all the chlorine and to limit the formation of hypochlorite (strongly corrosive) while caustic washing the liquid and gaseous effluents.

The effect of the mole ratio of C_2H_4 to Cl_2 on the degree of substitution is highly controversial. In certain cases the degree of substitution does not depend on the ratio of C_2H_4 to Cl_2 ; in other cases on the other hand the degree of substitution passes through a minimum for a very precise value of the ethylene to chlorine ratio.

At 50°C and with a circulation rich in $FeCl_3$ the degree of substitution does not vary while the mole ratio C_2H_4/Cl_2 varies between 0.93 and 1.14 (E.A.T. of Jemeppe June 1963). The same observations were made during the chlorination of strong ethylene with dilute chlorine (Note FN 1184) and of dilute ethylene by strong chlorine (9). Ebelin, de Pauw and Limido (20) state that the degree of substitution passes through a minimum for the mole ratio

$$\frac{C_2H_4}{Cl_2} = 2$$

(the degree of substitution obtained in this case being 4%). Ethyl feel that too high an excess of ethylene favours substitution. Galitzenstein and Woolf (40) mention that the degree of substitution is more sensitive to an excess of ethylene when the ferric chloride content of the circulation is low while Doroganew-kaja (cited by ref 40) mentions that it was necessary for him to use a mole ratio $C_2H_4/Cl_2 = 3$ to 4 to obtain very pure EDC.

Trials carried out at Jemeppe in the pilot plant (Jemeppe Note 437) shewed that the degree of substitution increased while the mole ratio C_2H_4/Cl_2 became less than 1. Galitzenstein and Woolf carried out the chlorination of ethylene by introducing stoichiometric quantities of the reactant into a circulation containing 5% w/w of chlorine and of ferric chloride. Under these conditions the degree of substitution was the same as when the circulation contained no chlorine.

1-Mole Ratio C_2H_4/Cl_2 less than or Equal to 1

The use of an excess of chlorine leads to an increase in the degree of substitution which is attributable to a prolonged contact of EDC with chlorine. However the substitution reaction can be practically arrested by adjusting the ferric chloride content. In industrial reactors (Ethyl, ICI, Solvay) the ferric chloride content (30-60 mg/kg) is not sufficient to stop completely the substitution reaction in the presence of an excess of chlorine.

2-Mole Ratio C_2H_4/Cl_2 Greater than 1

A slight excess of ethylene (1 to 2%) makes it possible to consume all the chlorine introduced into the reactor. However according to the addition and substitution mechanisms proposed in paragraph II A it appears that the order of the reaction with respect to ethylene is higher in the radical mechanism than in the polar mechanism. The degree of substitution must therefore increase with the excess of ethylene.

The concentration of ethylene is almost zero in the liquid phase during all the reaction (see II-A). It is only at the end of the reaction when the chlorine content of the circulation is low that the ethylene concentration in the liquid phase increases thus favouring the radical as opposed to the polar mechanism. It seems however that the radical reaction is very limited.

The overall rate of reaction depends on the rate of transfer of the ethylene from the gas to the liquid phase when the ethylene concentration in the liquid phase is practically zero. This is achieved when the chlorine concentration of the liquid phase is sufficiently high. At the end of the reaction and especially in the presence of an excess of ethylene the ethylene concentration of the liquid phase is no longer negligible; the solubility of the ethylene which depends on the temperature of the circulation can then influence the degree of substitution.

Industrially the excess of ethylene is estimated from the ethylene concentration of the residual (vent) gases (which can be measured continuously with a non dispersive IR analyser).

The optimum value for the ethylene content of the residual gases seems to depend on the type of direct chlorinator and on the rate of production. For Ethyl the optimum ethylene content of the vent gases is from 25 to 30% at Houston and from 8 to 10% at Baton Rouge (Notice MD294). At Tavaux while working at normal rates the degree of substitution does not seem to alter while the ethylene content of the vent gas varies from 10 to 30%. On the other hand at high rates the degree of substitution increases from 0.4 to 0.8% while the ethylene content of the vent gases increases from 20% to 30%.

These few examples reflect the complexity of the phenomenon. It seems that the type of nozzle and the position of the ethylene and chlorine nozzles in the reactor play a basic role (41). The velocity of the reactants at the level of the nozzles proposed by Ethyl is 30 mtrs/sec for ethylene and 24 mtrs/sec for chlorine (Notice MP295). The following values were adopted for Ethyl's new direct chlorinator at Houston: 42 mtrs/sec for chlorine and 52 mtrs/sec for ethylene (51).

E THE EFFECT OF THE DILUTION OF THE REACTANTS WITH INERTS

The presence of inerts in the reaction gases hinders the transfer of chlorine and ethylene into the liquid phase. In principle, if efficient contact is maintained the inerts present do not influence the course of the reaction. Kureha (9) carried out without difficulty the chlorination of a 'cracking' gas containing 13% ethylene: at 60°C and under a pressure of 4.2 kg/cm² the following results were obtained.

Ethylene converted	: 95%
Purity of EDC obtained	: 99%
Conversion of chlorine	: 100%

The use of very dilute gases was studied at NOH (Note FW1184): excellent conversions were obtained by carrying out the reaction in an irrigated scrubber.

For an industrial direct chlorinator working under normal conditions the presence of 5-10% v/v of inerts (referred to the volume of the reactants) cannot appreciably modify the degrees of conversion and of substitution.

Several years ago Ethyl chlorinated dilute ethylene (composition: C₂H₄ 30%, C₂H₆ 10%, CH₄ 40%, H₂ 70%) without encountering special difficulties. The β -tri content of the EDC was between 2 and 3% (Note PB834).

F THE EFFECT OF TEMPERATURE

In the complete absence of inhibitors (O₂ & FeCl₃) the degree of substitution is strongly dependent on temperature: 6% at -25°C, 29% at 0°C and 82% at 20°C (42).

In the presence of FeCl₃ the degree of substitution does not vary between 0 and 60°C. Several authors mention that above 60°C an increase in the degree of substitution is observed. From this temperature the vapour pressure of EDC increases rapidly and the enrichment of the vapour phase with EDC hinders the passage of ethylene towards the liquid phase. The dissolved chlorine thus remains in contact with the EDC for a longer period resulting in an increase in the degree of substitution. It should be remembered that the addition of sufficient FeCl₃ makes it possible to stop substitution reactions almost completely. Consequently when an increase in the temperature of the circulation is accompanied by an increase in the degree of substitution FeCl₃ must be added to bring the degree of substitution back to a reasonable level. The working temperature adopted by Ethyl is 70°C. It is not critical so long as the iron content of the medium is adequate. ICIANZ who also chose 70°C as the working temperature were obliged to add FeCl to keep the degree of substitution at a normal level (51).

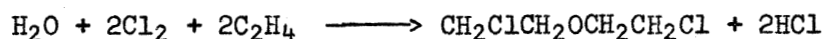
The favourable influence of FeCl_3 is exploited in chlorination under boiling conditions where the substitution reactions are controlled by the addition of FeCl_3 .

According to Ethyl, it is preferable for the circulation to be in the same direction as the reactants so that these are introduced at the coldest point of the reactor (Notice MP 294). However the direct chlorinators on VC3 work without any particular difficulty with the circulation counter current to the reactants.

G INFLUENCE OF WATER

1-Chemical Action of Water

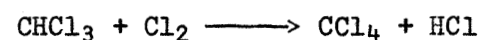
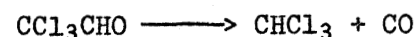
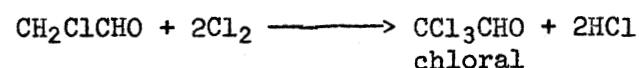
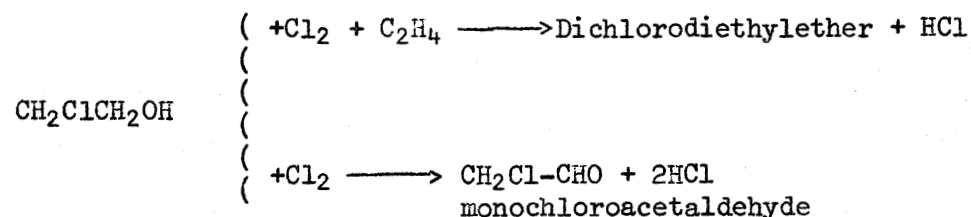
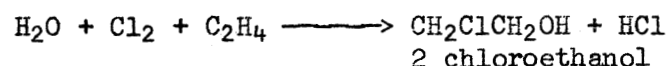
Water can enter the direct chlorinator either with the reactants or by leaks from the cooler placed in the circulation; the water reacts with the chlorine and the ethylene to form $\beta\beta$ dichlorodiethylether:



This reaction proceeds by a polar mechanism via 2 chloroethanol ($\text{CH}_2\text{ClCH}_2\text{OH}$).

When the chlorination of ethylene takes place in a glass apparatus at a temperature of $40-50^\circ\text{C}$ almost all the water introduced into the reaction medium is converted into the dichlorodiethylether. In the presence of a large amount of water 2 chloroethanol is also found (Note FN 1184).

In the case of chlorination under boiling conditions the water is entrained with the EDC vapour and consequently the formation of dichlorodiethylether is almost zero. However, it forms small quantities of chloroform and carbon tetrachloride which we attribute to the following reactions:



The chlorination of ethanol into chloral is a known reaction; it is carried out elsewhere industrially under conditions similar to the chlorination of ethylene (32).

2 chloroethanol and chloral are present in small quantities in the liquid phase. The transformation of chloral into chloroform and formic acid or carbon monoxide is easily achieved either by basic hydrolysis or by thermal decomposition. In the case of chlorination under boiling conditions it has not been possible to confirm by which mechanism (catalytic or non catalytic) chloral is transferred into chloroform. The mechanism must explain why in conventional chlorination the chloroethanol gives dichlorodiethyl ether while in chlorination under boiling conditions chloroethanol is converted to chloral. In conventional industrial chlorination which takes place at about 60°C traces of chloroform are also found in the EDC made.

2-The Influence of Water on Corrosion

The introduction of water into the reaction medium increases appreciably the corrosion of industrial direct chlorinators made out of mild steel from which the ferric chloride concentration of the circulation increases. The water does not disappear in the course of the corrosion; it catalyses the reaction between the steel and the chlorine giving ferric chloride. The water is eliminated from the reaction medium either in the form of dichlorodiethyl ether (or 2 chloroethanol) or with the EDC product. It follows that in the event of an instantaneous introduction of water, the rate of corrosion will depend not only on the quantity of water introduced with the reactants but also on the operating rates: at high rates the elimination of water will be achieved principally by the production of EDC while at low rates it will be by its conversion to dichlorodiethyl ether (conversion function of temperature).

The water introduced into the reaction finds itself in the presence of 3 hygroscopic agents: HCl , FeCl_3 and the metal surface. It can form different hydrated species such as $\text{H}_3\text{O}^+\text{Cl}^-$, $\text{FeCl}_3 \cdot \text{H}_2\text{O}$, $\text{M} \cdot \text{H}_2\text{O}$ etc In practice it is not possible to define the form in which the water is present in the chlorinator. The affinity of the water (or one of the hydrated species) for the mild steel wall appears to be sufficiently high for an appreciable increase in the rate of corrosion to be observed when the quantity of water introduced increases slightly. Normally the corrosion of mild steel by moist chlorine is accompanied by the formation of a protective film consisting of a mixture of $\text{FeCl}_3 \cdot 2\frac{1}{2} \text{H}_2\text{O}$ and of FeCl_3 (Bulletin PCT of 9. 12. 1966 and (33)). However due to the solubility of ferric chloride in EDC, this protective film is eliminated gradually as it is formed.

While the water content of the reactants remains low ($<20 \text{ mg/m}^3$) there is no marked corrosion in reactors and the dissolved iron content (built up) is in the region of 15-20 mg/kg. Above a limit still badly defined but in the region of $100 \text{ mg H}_2\text{O/m}^3$ the

rate of corrosion increases appreciably. For example 150 to 200 mg/m³ of moisture in the ethylene corresponds to a dissolved iron content in the liquid phase of 200 mg/kg. The liquid becomes blackish and serious difficulties are encountered with phase separation in the caustic wash (observations made at Tavaux).

Due to the removal of the water with EDC vapour (azeotrope), the rate of corrosion must be slightly higher in the part of the reactor which is in contact with the vent gases (especially in this part of the reactor the walls are continually washed with fresh EDC which condenses there and which contains dissolved Cl₂, HCl and O₂).

It should be mentioned that most industrial reactors described in the literature are made from mild steel.

As a result of what has been said above, the introduction of metallic surfaces into the reactor must therefore reduce its rate of corrosion of the reactor itself; this was effectively observed by Ethyl (34).

The rate of corrosion increases with the amount of free chlorine present in the liquid phase (Notices DGMP 294 and 295). Consequently any increase in the rate of chlorination of ethylene decreases the free chlorine concentration of the liquid phase and the rate of corrosion.

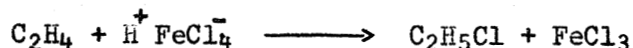
In the case of chlorination under boiling conditions laboratory trials shew that serious corrosion occurs in the condenser in the region where the EDC vapours condense. Due to the existence of an EDC water azeotrope these vapours are particularly rich in water. (See III.0).

H FORMATION OF ETHYL CHLORIDE

Ethyl chloride is formed by the addition reaction



This reaction which takes place in the liquid phase is catalysed by ferric chloride probably according to the following mechanism:



While the formation of ethyl chloride is a direct consequence of the formation of HCl from secondary reactions, the quantity of ethyl chloride formed depends principally on the presence of ferric chloride in the reaction medium. The results given below are for conventional

chlorination; they have been confirmed for chlorination under boiling conditions. In this case when the reaction is carried out in a circulation of perchloroethylene (see III-H-2) where the ferric chloride content is of the order of 1 mg/kg, the quantity of ethyl chloride collected in the vent gases is very small (<0.05 g/kg). On the other hand when the ferric chloride content is from 2-3 g/kg (in a circulation of EDC) the quantity of ethyl chloride formed is much higher.

Mole ratio C_2H_4/Cl_2 : 1.05 Circulation temperature: 45°C Circulating medium: EDC			
Fe Cl_3 Mg/ltr	O ₂ % of vol C_2H_4	Degree of substitution %	Ethyl Chloride g/kg of circulation
Traces	0	28	0.2
Traces	2	2.7	<0.05
250	0	1.89	2.8
250	2	0.08	1.2

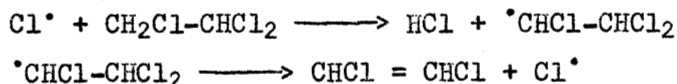
When the degree of substitution is the order of a few percent (absence of oxygen and small quantities of ferric chloride) the amount of ethyl chloride formed is considerable. This explains what was observed in the pilot plant at Jemeppe when the chlorine was introduced into the EDC circulation before the ethylene (Jemeppe Note No 437).

When the action of ferric chloride present in the reaction medium is neutralised by the addition of gaseous ammonia (probably with the formation of the $FeCl_3 \cdot NH_3$ complex) the ethyl chloride content of the circulation and of the vent gases becomes almost zero inspite of the very considerable increase in the HCl content (see the effect of ammonia II-L).

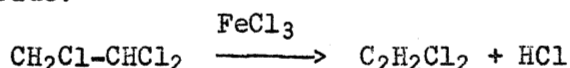
I FORMATION OF DICHLOROETHYLENES, TETRACHLOROETHANES AND OF HIGHER CHLORINATED COMPOUNDS

As we will see below these compounds appear mainly during chlorination under boiling conditions; their formation depends on the contents of β -tri and ferric chloride in the circulation as well as the temperature.

The 1,1 as well as the cis- and trans-1,2-dichloroethylenes are formed by the dehydrochlorination of β -tri; the 1,2-dichloroethylenes (especially the cis isomer) are obtained principally. The dehydrochlorination of β -tri is not carried out according to a radical mechanism such as



It is rather a monomolecular dehydrochlorination catalysed by ferric chloride:



In practice it has been shown that a $\text{Cl}^\bullet$ radical reacts more slowly with β -tri than with EDC (28). For a radical mechanism the quantity of HCl liberated should fall while the circulation becomes richer in β -tri. But this is contrary to what is observed. Furthermore in the absence of chlorine (ie in the absence of any possibility of reaction with $\text{Cl}^\bullet$) while purging through with air a reactor kept boiling and containing about 50% w/w β -tri amounts of HCl comparable with those observed while chlorine is being introduced are liberated (See III-H).

In the presence of chlorine cis and trans dichloroethylenes are converted into 1,1,2,2-tetrachloroethane.



There is also formation of a small amount of 1,1,1,2-tetrachloroethane resulting from the chlorination of 1,1-dichloroethylene.

In a recent patent the manufacture of 1,1,2,2-tetrachloroethane by the chlorination in the liquid phase of 1,1,2-trichloroethane in the presence of AlCl_3 is claimed (29). The reaction probably takes place by means of a dehydrochlorination mechanism identical to that described above.

Trichloroethylene, perchloroethylene, pentachloroethane and hexachloroethane are also formed by a similar mechanism.

The formation of dichloroethylenes and tetrachloroethanes is practically zero in conventional industrial chlorination both at Ethyl and at Tavaux. This is due to the fact that in the reactor:

- β -tri is only present in small amounts.
- The concentration of ferric chloride is low.
- The temperature is lower than in chlorination under boiling conditions.

J FORMATION OF C₄ DERIVATIVES

C₄ derivatives formed in parallel to EDC originate from the reaction of C₂ radicals with ethylene or with themselves. The C₄ compounds can then be isomerised in the presence of FeCl₃ and react with HCl and Cl₂. C₄ compounds detected at Tavaux are 1,2-dichlorobutane, 1,3-dichlorobutene-2, 1,2,3-trichlorobutane and 1,2,4-trichlorobutane. These compounds only represent 0.05% w/w of the EDC manufactured (Notice DT/NOH-P 68.338).

1,2-dichlorobutane could come from the chlorination of 1 butene. This is one of the impurities of ethylene, but the amount present is too small to account for all the 1,2-dichlorobutane formed. It is not impossible that very small quantities of 1-butene are formed by the dimerisation of ethylene in contact with FeCl₃.

K FORMATION OF VARIOUS COMPOUNDS FROM IMPURITIES IN THE REACTANTS

Compounds such as 1-bromo-2-chloroethane, β-β-dichlorodiethyl-ether and chloroform are due to impurities in the reactants. The mechanism of their formation is discussed below when the effect of impurities is being examined.

L EFFECT OF IMPURITIES IN THE CHLORINE

Chlorine currently used at Tavaux (Chlorine A) contains the following impurities.

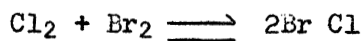
N ₂	0.3-1.1% v/v
H ₂	1.3-1.8% "
CO ₂	0.3-0.7% "
O ₂	0.3-1.1% "
H ₂ O	<10-50 mg/m ³

Nitrogen, hydrogen, and carbon dioxide behave as inert gases and have no direct influence on the chlorination (see II E).

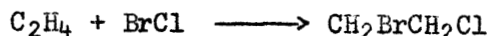
The effects of water and oxygen have already been described in detail earlier (II-C and G).

Chlorine A also contains very small quantities of Br₂, NCl₃, COCl₂, CH BrCl, CHCl₃, CCl₄, C₂Cl₆, C₆Cl₆ and perhaps NOCl.

Bromine which is the most abundant impurity (80-100 ppm w/w) turns up in the form of 1-bromo 2-chloroethane in the EDC formed. In the presence of chlorine an equilibrium exists:



Br Cl is much more reactive than Cl₂ (35) and reacts with ethylene:



Among the other impurities only NCl_3 and NOCl could influence the formation of β -tri (radical reaction). Bearing in mind that they are only present in the chlorine in extremely small amounts this influence is negligible.

At normal temperatures and probably up to 100°C nitrogen trichloride in dilute solution does not generate free radicals liable to accelerate the formation of β -tri. On the contrary NCl_3 rather behaves here as a capturer of radicals:



The reverse reaction is very slow at ambient temperatures (36) $\text{NCl}_2^\bullet$ is finally transformed into N_2 and Cl_2 (37).

M THE EFFECT OF IMPURITIES IN THE ETHYLENE

The purity of ethylene is normally greater than 99% v/v. The impurities in ethylene are normally

- methane, ethane, higher olefins
- very small quantities of H_2 , O , CO , CO_2 , NO , H_2O and acetylenes
- occasionally ammonia and methanol

Methane and ethane behave as inert gases.

Chlorine and ethylene are mixed from their introduction into the reactor; consequently the H_2 , O_2 , CO , CO_2 and H_2O have the effects already described in the case of their introduction with the chlorine.

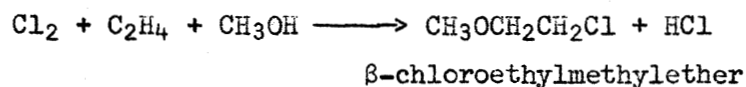
Olefins and acetylenes are converted into the chlorine addition compounds: propylene into 1,2-dichloropropane, butadiene into 1,2,3,4-tetrachlorobutane, acetylene into 1,1,2,2-tetrachloroethane.

The effect of NO is comparable to that of NCl_3 and NOCl introduced with the Cl_2 .

The particularly marked deleterious effect of ammonia was demonstrated in the plant at Tavaux while using ethylene containing about 100 ppm of this impurity. The ammonia neutralizes the action of dissolved FeCl_3 : the complex $\text{FeCl}_3 \cdot \text{NH}_3$ is formed with the result of a violent increase in the degree of substitution (which approached the degree observed in the laboratory while carrying out chlorination with oxygen on its own as inhibitor for the substitution reaction). At the same time the colour of the reaction medium passed from very dark brown to a very clear yellow. A light precipitate is sometimes detected. The addition of a sufficiently large supplementary amount of FeCl_3 to the reactor circulation makes it possible to eliminate the deleterious influence of the ammonia and to return the degree of substitution to its normal value.

	Ethylene without NH ₃	Ethylene with NH ₃
REACTOR CIRCULATION		
HCl, mmol/ltr	approx 10	40-50
β-tri, % w/w	0.15-0.3	1.7-2.5
C ₂ H ₅ Cl, ppm w/w	180-300	<10-20
Iron, mg/ltr	25-35	25-35
REACTOR VENT GAS		
HCl, mol %	2.7	20
C ₂ H ₅ Cl, mol %	0.23	<0.01

In the presence of HCl methanol can give methyl chloride and water. However like water methanol can in the presence of chlorine and ethylene give a chloroether:



REMARK

Propylene is not harmful for chlorination; on the other hand the 1,2-dichloropropane formed is a poison for cracking (38). Since it is difficult to eliminate the dichloropropane by distillation it is necessary to limit the propylene content of the ethylene. In the absence of exact data on the effect of the dichloropropane on the cracking of EDC the propylene content of the ethylene is currently limited to 50 ppm w/w.

N THE EFFECT OF HCl

The addition of gaseous hydrogen chloride into the reactor seems to lead to an increase in the degree of substitution. Thus for example the introduction of 0.0208 mol HCl per kg of EDC increases the percentage of β-tri in the EDC from 0.05 to 0.28% in the experimental conditions described in the table of chapter II C).

Ethyl also observed industrially that the introduction of HCl into the direct chlorinator was accompanied by an increase in the β-tri content of the EDC manufactured (39). It is possible that the HCl reacts with FeCl₃ to form the complex FeCl₄H⁺ which does not sufficiently inhibit the radical reactions leading to the formation of β-tri (see II-B).

0 CRITICAL FACTORS WHICH GOVERN THE COURSE OF THE REACTION IN AN INDUSTRIAL DIRECT CHLORINATOR

When chlorine and ethylene are introduced into the liquid EDC, the chlorine dissolves rapidly while the gas phase becomes concentrated in ethylene. Simultaneously the reaction between the chlorine and the dissolved ethylene occurs. Due to the high velocity of the reaction in the liquid phase, the ethylene content of the liquid phase becomes practically zero; the reaction takes place at the gas liquid interface towards which the chlorine in the liquid phase and the ethylene in the gas phase migrate. The reaction rate is thus limited by the rate of diffusion of ethylene to the interface (see II A).

In the bulk of the liquid phase a fraction of the dissolved chlorine reacts with the EDC to form β -tri. The substitution reaction initiated at the interface by ethylene (see II A) is inhibited by ferric chloride and dissolved oxygen.

Let us examine in detail the behaviour of a gas bubble consisting initially of a mixture of chlorine and ethylene: due to the high solubility of chlorine in EDC, the ethylene content of the gas phase is higher than the chlorine content. At the gas liquid interface a very rapid reaction occurs between the chlorine and the ethylene which leads to the chlorine content at the periphery of the bubble being practically zero. Since the homogenization of the gaseous mixture is slow, the chlorine present at the centre of the bubble is entrained in a shell of ethylene to the surface of the liquid. This mechanism explains why in the absence of sufficient effective contact between the gas and liquid phases chlorine is found at the end of the reaction in the vent gases even when there is excess ethylene.

It is therefore not necessarily under sizing of the reactor which is responsible when the reaction is incomplete but more often a lack of effectiveness due to insufficient agitation.

At normal rates, Ethyl (Houston Reactor) find 6-8% w/w chlorine in the vent gases; at low rates the chlorine content does not go below 2.75% w/w (43).

These difficulties are not encountered at Tavaux (see schedule in Table 1 in the appendix) nor at ICI-ANZ (44) and AE and CI (45).

The form, distribution and number of nozzles are the main factors for ensuring efficient contact between the gas and the liquid phases.

When, in a reactor which is functioning correctly, a sudden increase is observed in the chlorine content of the vent gases, it is necessary to check that there is neither blockage nor corrosion of the nozzles.

In the case of a boiling reactor there is vaporisation of EDC at the gas liquid interface of the bubbles which restricts the diffusion of ethylene into the liquid. Compared with conventional chlorination, the chlorine remains longer in contact with the EDC which increases the risk of forming β -tri. It is necessary to increase appreciably the ferric chloride content of the liquid phase to limit the substitution reaction.

III CHLORINATION OF ETHYLENE IN A LIQUID MEDIUM MAINTAINED AT ITS BOILING POINT

A REVIEW OF PATENTS ON CHLORINATION UNDER BOILING CONDITIONS

In one of its patents Union Carbide proposed carrying out the chlorination of ethylene under boiling conditions (13). The reactor 2.1 mtrs dia and 13.1 mtrs high is fitted in its upper part with 16 bubble trays 30 cm apart. The reactor contains about 150 t of EDC to which is added a little ferric chloride. The ethylene and the chlorine are introduced in a mole ratio of 1.2 to 1 into the boiling EDC. The heat of reaction is controlled by the reflux ratio which is about 5 (temperature of EDC, 90°C pressure 0.5 atms gauge). The pure EDC is removed from the third tray from the top and is used as such for cracking (HCl content 23 ppm; moisture content 34 ppm). The β -tri which builds up in the reactor can be removed either by taking off from the bottom tray or by purges from the base of the reactor.

In a very similar patent, Columbia Southern (11) does not remove the heavies by purges from the reactor; it is therefore necessary to remove its liquid contents approximately every six months. The reactor is not surmounted by a rectification column and the process only envisages obtaining EDC free from FeCl_3 ; the EDC produced is separated from light and heavy constituents by two subsequent distillations.

The patents of Dynamit Nobel (12) and Chemische Werke Hüls (14) give variations of the two processes described above.

Kureha (9) uses the boiling reactor technique with dilute ethylene originating from cracking gas. The reaction temperature is fixed at 60°C which makes it possible to remove the EDC produced in the form of a vapour with the cracking gases not consumed in the reaction.

The search carried out by Service Brevets (bulletins LM/AD of 22.8.69 and LM/PK of 25.11.68) concluded that we are free to use the process for the chlorination of ethylene under boiling conditions with rectification in the following countries: Belgium, Spain and Brasil. The process will be 'public' in 1970

in Italy and 1975 in France. No corresponding Union Carbide patent or request for a patent was found in Germany or Austria.

B ETHYL'S EXPERIENCE

Ethyl have built at Thessaloniki (Greece) a boiling reactor with a capacity of 30,000 t of EDC per annum. The construction of the reactor was preceded by two summary studies carried out in the laboratory (46). These studies shewed that it was possible to carry out the chlorination of ethylene under boiling conditions without appreciable lowering of the yield.

The reactor was designed so that it could be easily converted into a conventional chlorinator (47). Part of the EDC leaves the chlorinator as a liquid so as to form a continuous purge. The vaporised EDC which represents about 85% of the production is used to make vinyl chloride (cracking) while the fraction removed as a liquid undergoes the conventional treatment of caustic washing and is used for the preparation of additives for petrol.

The results obtained so far to not make it definitely possible to opt for chlorination under boiling conditions. More especially the ethylene consumption is 10% higher than for conventional direct chlorinators and the purity of the EDC obtained is lower (48).

After some weeks of operation all the EDC produced was taken off in the vapour phase so as to maintain a ferric chloride concentration in the circulation sufficiently high and to lower the formation of β -tri.

In spite of the difficulties encountered in Greece Ethyl have mentioned in their monthly reports their intention of constructing a boiling reactor at Houston, the detailed design of which is now completed. As a result of local economic considerations, Ethyl has meanwhile withdrawn its proposal for the moment. (Report for March 1969).

C PILOT PLANT TRIALS AT JEMEPPE

During the course of a preliminary study carried out at Jemeppe trials were carried out under reduced pressure with a view to making use of the heat of reaction to vaporise the EDC and so obtain it rectified and free from FeCl_3 and HCl . At a pressure of 160 mm of mercury and a temperature of 380°C , the flows of chlorine and ethylene which at atmospheric pressure were 20 mol/hr had to be reduced to 10 mol/hr to avoid losses of C_2H_4 and Cl_2 .

Under these conditions the degree of substitution was appreciably higher (E.A.T. Jemeppe, April 1963).

These trials were not pursued.

D COMPARISON BETWEEN THE CONVENTIONAL PROCESS AND THE BOILING PROCESS

Caustic washing of the EDC obtained by conventional chlorination encounters serious difficulties when the ferric chloride content of the EDC manufactured exceeds a certain figure: ferric hydroxide floc accumulates in the caustic separator and make the separation between EDC and the caustic solution difficult. After this treatment the wet EDC is dried by azeotropic distillation. The quantity of ferric chloride necessary for satisfactory operation of the chlorinator is close to the limit mentioned above. In certain cases it is necessary to make a compromise taking into account the difficulties encountered with phase separation and the optimum FeCl_3 content to maintain in the reaction medium.

The removal in the vapour state of the EDC produced makes it possible to obtain a product practically free from FeCl_3 which no longer has to be caustic washed; lights such as chlorine and HCl can be eliminated by desorption. This way of operating makes it possible to adjust the ferric chloride content to the optimum value and to obtain EDC purified from heavies.

Three techniques can be used to remove the EDC produced in the vapour phase:

- 1 Operate below the boiling point and carry over EDC vapour with inert gas introduced into the reactor with the reactants. This technique was used by Kureha (9) for the chlorination of dilute ethylene resulting from the cracking of naphtha. The yields in this case are slightly less than those from conventional chlorination with undiluted ethylene.
- 2 Operation at the boiling point at reduced pressure. This technique makes it possible to adjust the temperature by modifying the operating pressure. To our knowledge this technique has not been used industrially; it was the object of a very brief study in the 'Test Hall' at Jemeppe (See III C). Also a pilot plant (with a capacity) of about 100 kg/day worked with excellent yields at the Shri Ram Institute for Industrial Research (5).
- 3 Operation at the boiling point at atmospheric pressure: in conventional chlorination increasing the temperature above 60°C is generally accompanied by an increase in the degree of substitution. While working under boiling conditions at atmospheric pressure it is therefore necessary to keep the degree of substitution at an acceptable level. This is achieved by the addition to the reaction medium of sufficient FeCl_3 . According to published patents chlorination under boiling conditions at atmospheric pressure seems to be carried out in several industrial plants; one is limited however in the patents on purification by vaporisation of the EDC produced. For our part we have studied chlorination under boiling conditions while

envisaging the use of the heat reaction to purify by vaporisation not only the EDC from direct chlorination but also the EDC from oxychlorination and/or the EDC recycled from cracking, these streams being introduced into the boiling reactor.

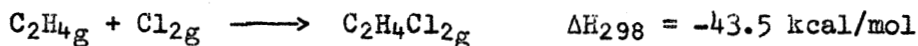
E THE AMOUNT OF HEAT RECOVERABLE IN AN INDUSTRIAL PLANT CONSISTING OF A COMBINATION OF CHLORINATION, OXYCHLORINATION AND CRACKING

BASIS

The gaseous ethylene and chlorine are supplied at 75°C. It is not necessary to reheat EDC from oxychlorination since the EDC recycled from cracking is available before let down at 155°C and that consequently the mixture can be introduced into the chlorinator at 83°C.

THERMAL BALANCE

Heat of reaction



Heating of the EDC vapour from 25 to 83.5°C:

$$0.0188 \times (83.5 - 25) = 1.1 \text{ kcal/mol}$$

Heat to be removed

$$51.3 - 1.1 = 50.2 \text{ kcal/mol EDC produced by chlorination}$$

Symbols (for 1 mole of VC produced)

C: Direct chlorination EDC, mole/mol VC

l-c: Oxychlorination EDC, " " "

l: Total EDC

r: reflux ratio

x: depth of crack

Heat to be removed: $50.2c$ kcal/mol VC

Latent heat of vaporisation of EDC at its boiling point:
7.8 kcal/mol.

	EDC to Vaporise	Heat Required
Chlorination	C	7.8c
Oxychlorination	1-C	7.8 (1-c)
EDC recycle from cracking	$\frac{1}{x} - 1$	$7.8(\frac{1}{x} - 1)$
Total	$\frac{1}{x}$	$7.8/x$
Reflux	r/x	$7.8 r/x$
Total	$\frac{1+r}{x}$	$7.8 \frac{1+r}{x}$

The heat to be removed as a function of c is given in diagram 2A.
The heat of vaporisation of EDC as a function of the reflux for different values of x is given in diagram 2B.

Heat which can be recovered under the operating conditions at Tavaux

As an example Tavaux's operating point (according to the flow sheet) is given in diagram 2: C = 0.39 x = 0.5 r = 0.77

In this case one has

Heat to be removed: 19.6 kcal/mol VC
Heat required: 27.6 kcal/mol VC

One can therefore recover all the heat of reaction. Additional heat to supply

$$27.6 - 19.6 = 8.0 \text{ kcal/mol VC}$$

If one expresses the quantities of heat to be removed and required as kg of steam per kg of VC produced one has:

- steam removed $\frac{19.6 \times 16}{530} = 0.59 \text{ kg steam/kg VC}$
- steam required $\frac{27.6 \times 16}{530} = 0.83 \text{ kg steam/kg VC}$
- additional steam: $0.83 - 0.59 = 0.24 \text{ kg steam/kg VC}$

Effect of working conditions on heat recovered

It can be conceded that the actual working conditions can be encompassed between the following limits:

$$r = 0.6 \text{ to } 1$$

$$x = 0.5 \text{ to } 0.65$$

$$c < 0.5 \text{ (generally due to HCl originating from the manufacture of chlorinated products)}$$

The minimum heat necessary for the vaporisation of EDC corresponds to $r = 0.6$ and $x = 0.65$; under these conditions 19.2 kcal/mol of VC are required. For $C = 0.39$, 19.6 kcal/mol of VC have to be removed; a reflux ratio of 1.1:1 can be allowed when $x = 0.65$.

F TEST APPARATUS (Scheme 3)

Preliminary trials were carried out without effective rectification of the EDC product which is shown by the entrainment of β -tri with the EDC product. With a view to improving the purity of the EDC with respect to β -tri and heavier components, the gases leaving the reactor were treated in a glass rectifying column 50 mm dia, 3950 mm high and packed with 10 mm x 10 mm glass Raschig rings. Heavies accumulating in the boiler were recycled to the reactor by means of a pump.

The chlorinator is a vertical Pyrex tube (diameter 50 mm height 4660 mm) connected to a tube (diameter 25 mm, height 3260 mm) which ensures rapid circulation of the liquid contents by thermosyphoning and by the mechanical action of the gas bubbles. The chlorinator and the thermosyphon are provided with jackets for the circulation of thermostatically controlled paraffin oil. For most of the tests the reactor was packed with 10 mm x 10 mm mild steel Raschig rings for a height of 1650 mm. The whole apparatus is covered with red adhesive ribbon, transparent but opaque to actinic light, which makes it possible to detect rapidly any malfunctioning while avoiding reactions induced by light. The EDC is purified from lights and more especially chlorine and hydrogen chloride in an adiabatic Vigreux with 408 points (24 mm dia., 1100 mm high).

The ethylene and chlorine air mixture are introduced separately into the base of the chlorinator. The flows are measured by rotameters kept at a pressure of 1292 mm Hg with the aid of pneumatic regulation. For some trials the chlorine and ethylene were mixed in a tee 15 cm before being introduced into the reactor.

After reaction the vent gases pass successively through a chlorine analyser, a scrubber irrigated with a dilute solution of caustic soda, an ethylene analyser and a gas meter. For some trials the gases were also passed through an oxygen analyser. The analysis techniques used are described in note FN 1184.

G METHOD OF OPERATION

- The reactor is filled with EDC up to the level of the thermosyphon.
- The EDC contents are heated to boiling point and the distillation columns put on line.
- Dilute caustic is circulated through the scrubber.
- Air is passed first of all through the reactor followed by chlorine and ethylene.
- The reaction starts up instantaneously and the start up of the reactor is very rapid.
- The reflux and the off takes from the distillation columns are adjusted to working conditions with the aid of rotameters.
- The residual chlorine and HCl are titrated from the liquid collected at the scrubber. The chlorine content (generally zero) makes it possible to follow the progress of the reaction while the HCl content makes it possible to measure the degree of substitution.

H THE EFFECT OF ORGANIC COMPOUNDS IN THE CIRCULATION

1-The effect of β -tri

Initially the EDC introduced into the reactor is 99.9% w/w $C_2H_4Cl_2$. The degree of substitution is initially a few %; it falls very rapidly to stabilize at about 0.2 to 0.6% provided that the iron content of the circulation is greater than 0.2-0.3 g/kg (or 0.6-0.9g $FeCl_3$ /kg). So long as the EDC content of the circulation is greater than 85% w/w, the degree of substitution rarely exceeds 0.6% (table 4 in the appendix and tables p 36, 37, 38 and 40).

In the case of the trials which are the subject of Table 4, the chlorine and ethylene were mixed in a tee, 15 cm before the introduction of the gases into the liquid. When the gases are introduced separately, the degree of substitution is slightly higher and lies between 0.4 and 0.7%.

The addition of β -tri to the EDC of the circulation so as to make it 50-60% of the mixture is accompanied by an increase in the degree of substitution calculated from the HCl liberated.

The apparent increase in the degree of substitution depends on the $FeCl_3$ and β -tri contents in a complex manner. In practice one part of the $FeCl_3$ inhibits the radical substitution reaction which forms the HCl and the β -tri and the other part of the $FeCl_3$ catalyses the dehydrochlorination of the β -tri. Moreover the addition of the β -tri to the circulation increases the boiling

point from 80-81°C to 94-95°C which further favours the dehydrochlorination reactions catalysed by the FeCl_3 .

It is difficult in our conditions to evaluate the real degree of substitution from the composition of the circulation. It is on the other hand possible to follow the dehydrochlorination reaction of the β -tri by bubbling air through the circulation raised to its boiling point. The quantity of HCl liberated while doing this is of the same order of size as during the normal operation of the reactor. Since the rate of substitution of an atom of hydrogen by an atom of chlorine is less in the case of β -tri than in that of EDC (see II-I) one can say that the real degree of substitution is not modified by the addition of β -tri to the circulation. If the quantity of ethylene converted into β -tri does not alter, the consumption of chlorine, in contrast, increases because there is chlorination of the cis and trans 1,2-dichloroethylenes formed by the dehydrochlorination of β -tri into 1,1,2,2-tetrachloroethane. (See Table V in the appendix).

For run No.57 the results are averages established over a duration of 36 hours while for runs 46 to 49 the measurements represent averages over 2 hours. During run 57 the chlorine and the ethylene were mixed just before their introduction into the reactor. In run 49 the flow of chlorine was estimated from the ethylene content of the vent gases.

2-Perchloroethylene Circulation

The β -tri is formed by the reaction of chlorine with EDC according to a radical mechanism (see II-B). If the EDC formed by the addition of chlorine to ethylene was eliminated as it was formed the quantity of β -tri formed should necessarily decrease appreciably. From this one can envisage the replacement of the EDC circulation by a solvent which would be inert with respect to chlorine and possess a boiling point higher than that of EDC; in this way the EDC could be removed continuously by distillation.

Polyhalogen derivatives not containing hydrogen atoms reactive to $\text{Cl}\cdot$ radicals could be suitable as a solvent.

Bromine and iodine compounds are not suitable because the bromine and iodine could be substituted by the chlorine.

Among the fluorinated, chlorinated or chlorofluorinated derivatives, pentachlorofluoroethane, boiling at 135°C could be suitable providing it does not form an azeotrope with EDC; its melting point is however very high (100°C).

The tetrachlorofluoroethanes boiling at 90-93°C form azeotropes with EDC and are consequently unsuitable (BHIT No 52 p 65 1968).

Instead of preparing pentachlorofluoroethane we carried out a series of trials with a circulation of perchloroethylene. We were thus able to verify that by replacing EDC with an inert solvent it is in practice possible to maintain a degree of substitution of 0.1 to 0.3% with dissolved iron content not exceeding 0.002g/kg (see Table below). However under the conditions of chlorination, perchloroethylene is not completely inert with respect to chlorine; it is transformed slowly into hexachloroethane and is therefore not useable industrially. It should be mentioned that inspite of the high temperature of the circulation (110-113°C) the degree of substitution remains very small. The quantity of hexachloroethane produced represents approximately 3% of the EDC produced.

THE EFFECT OF USING A CIRCULATION OF PERCHLOROETHYLENE ON
THE DEGREE OF SUBSTITUTION

Run No		41	42	40	43	44
Flows Moles/hr	Cl ₂	6	10	20	20	20
	C ₂ H ₄	6.18	10.3	20.6	20.6	20.6
	Air	0.12	0.2	0.4	0.4	0.4
Average temperature of the circulation °C		112	113	98	112	102
Degree of substitution %		0.3	0.2	0.2	0.1	0.2
C ₂ H ₄ in the vent gas (±5)%		51	72	57	57	52
Composition of the Circulation g/kg	Fe	0.002	0.001	0.001	0.001	0.001
	EDC	28	25	61	30	13
	Perchloroethylene	931	918	884	892	776
	Hexachloroethane	37	53	51	72	202
	Other chlorinated derivatives		Balance			

I THE INFLUENCE OF IRON AND OXYGEN

The effect of ferric chloride and oxygen on the formation of secondary products has already been examined in detail from the point of view of mechanism in chapter II C; this aspect of the question will not be reproduced here.

In chlorination under boiling conditions when the circulation consists of EDC or a mixture of EDC and β -tri, it is essential that the dissolved iron content exceeds 0.3 gm/kg to keep the degree of substitution at an acceptable level (<1%). Trials carried out by Ethyl on a laboratory scale (46) and in the industrial reactor in Greece and also results published by Union Carbide (12) and Columbia Southern (10) confirm the necessity of maintaining a high ferric chloride content in the reaction medium to prevent the formation of β -tri.

When the reactor contains a metal packing it is not however necessary to add FeCl_3 to the circulation. The latter is formed in situ by the action of chlorine and the traces of water contained in the reactants on the steel. The degree of substitution initially high falls rapidly and stabilizes after a few hours at a value close to that obtained in conventional chlorination. This evolution of the degree of substitution shews that the metal on its own does not influence the reaction directly as supposed by Ethyl (31).

The presence of oxygen (air) in the reactants seems to be less critical for chlorination under boiling conditions than in conventional chlorination. The effect of oxygen is probably marked by the presence of large amounts of ferric chloride. In the presence of ferric chloride, the degree of substitution practically does not increase when trials are carried out in the absence of a metal packing and of oxygen.

J THE EFFECT OF THE $\text{C}_2\text{H}_4/\text{Cl}_2$ RATIO (See Table P 36)

Any excess of chlorine (in relation to ethylene) is indicated by the presence of chlorine in the vent gases. In conventional chlorination part of this excess of chlorine remains dissolved in the circulation and reacts to give β -tri. On the other hand in a boiling reactor the chlorine is removed at the same time as the EDC vapour and the inerts. The chlorine thus removed can react with the EDC (free from FeCl_3) in the rectifying column and form β -tri. (The glass column used in the laboratory was protected from the action of light - see run no 55 in the table on p. 36). Under our experimental conditions it was difficult to distinguish between substitution reactions taking place in the reactor and those taking place eventually in the rectifying column. The laboratory results obtained by Ethyl are similar to those obtained by us (46).

A 10% excess of ethylene above the stoichiometric requirement is accompanied by a slight increase in the degree of substitution (0.5% - runs 56 and 31 in the table on p 36). This result is in agreement with the observations made for conventional chlorination.

In its laboratory trials on boiling reactors Ethyl did not however observe an increase in the degree of substitution with an excess of ethylene. It should however be remarked that the average degree of substitution which Ethyl mentioned for their trials was of the same order of size as that which we observed with 10% excess ethylene (1.1%).

The presence of excess ethylene in the vent gases should be interpreted with care: in certain cases the excess ethylene is only a result of a high degree of substitution. In effect when due to insufficient inhibition of substitution, part of the chlorine reacts to form β -tri, the ethylene content of the vent gases must increase. A high ethylene content of the vent gas does not therefore necessarily constitute an indication that there is an abnormal excess of ethylene in relation to the chlorine. In the extreme case it is even possible to have an ethylene shortage at the inlet to the reactor (ethylene shortage in relation to chlorine) when substitution is considerable.

Under the optimum working conditions of the reactor minimum substitution occurs with an excess of from 0.5 to 2% ethylene both for conventional and boiling chlorination.

During our tests the ethylene content of the vent gas was monitored with the aid of an IR ONERA non-dispersion type 50 analyser. The measured content rarely remained constant; it varied between 20 and 60% v/v ethylene. In practice the effect of fluctuations $\pm 1\%$ in the values indicated by the rotameters corresponded to fluctuations of 10-20% in the ethylene content of the vent gas. Nevertheless, the ethylene content was normally kept reasonably constant during tests. In the laboratory it seemed that optimum working conditions in the chlorinator corresponded to an ethylene content in the vent gas of between 40 and 60% which is slightly higher than in a conventional chlorinator for which this content varies between 20 and 30% v/v.

It could therefore be worth considering having an auxiliary chlorination of vent gas alongside the boiling reactor.

NOTE

Industrially the excess ethylene is adjusted by following the ethylene content of the vent gas. Due to the method of measurement used, errors in this value can be caused by the presence in the ethylene of inert hydrocarbons such as methane and ethane. For example, a content of 1% ethane in the ethylene corresponds to an excess of 1% ethylene. Consequently when the ethylene used contains a few per cent of inert hydrocarbons, it is essential to ensure that the IR analyser is not sensitive to these impurities.

EFFECT OF C_2H_4/Cl_2 RATIO

Test No		55	31	56
Flows Moles/hr	Cl_2	20	20	20
	C_2H_4	19.8	20.6	22
	air	0.4	0.4	0.4
Mole Ratio C_2H_4/Cl_2		0.99	1.03	1.10
Average reactor temperature °C		82	84	82
Degree of substitution		1.4	0.6	1.1
C_2H_4 in the Vent Gas (± 5)%		<5	35-40	>80
Composition of Reactor Contents	Fe	1.5	1.2	1.5
	EDC	896	937	888
	β -tri	70	55	77
	Other chlorinated compounds Balance			
Flow of chlorine in the vent gas mole/hr		0.0125	0	0

K THE EFFECT OF DILUTION OF THE REACTANTS WITH INERT GASES

The dilution of the reactants with air up to 20 mol%, (referred to the chlorine) practically does not effect the reaction (see Table page 37). While operating at 10 moles of chlorine per hour there is an increase in the degree of substitution 0.5%, however, while operating at 20 moles of chlorine per hour the degree of substitution remains practically unchanged, while the amount of air varies from 2 to 20 mole % of the chlorine.

For practical reasons air was used as the diluent gas. The complete replacement of air with nitrogen however does not modify the course of the reaction. (It should be noted that in these tests the iron level in the contents of the reactor was very high (2.3 g/kg). It is likely that gases such as carbon dioxide, carbon monoxide and hydrogen present in the chlorine and methane and ethane probably present in the ethylene can be considered as inerts from the point of view of the reaction. (See II.E).

When the chlorine and the ethylene are injected simultaneously into the reactor it does not matter whether the ethylene or the chlorine contains the inerts, all other things being equal. When the reaction takes place in an irrigated scrubber where the reactants are injected at different levels (note FN.1184) the dilution of the ethylene is less favourable.

EFFECT OF DILUTION OF REACTANTS WITH INERT GASES

Test No		33	34	29	36	31	37	38
Flows moles/hr	Cl ₂	6	6	10	10	20	20	20
	C ₂ H ₄	6.18	6.18	10.1	10.3	20.6	20.6	20.6
	air	0.12	1.2	0.2	2	0.4	4	0.4(N ₂)
Average Reactor Temperature °C		82	83	81	82	84	73	82
Degree of Substitution		0.5	0.4	0.4	0.9	0.6	0.5	0.5
C ₂ H ₄ in the Vent Gas (± 5), %		48-63	16-23	25-55	11-15	35-40	20-23	47-56
Composi- tion of Reactor Contents g/kg	Fe	0.7	0.8	0.8	1.3	1.2	2.3	2.3
	EDC	941	940	967	934	937	928	905
	β -tri	51	51	26	56	55	60	76
	Other chlorinated compounds	Balance						

L THE EFFECT OF GAS FLOWS

In the range of flows studied (6-30 moles of chlorine per hour) no change was observed in the degrees of conversion and substitution (see Table, Below). The reaction is thus very rapid which is in agreement with Ethyl's laboratory results. The effect of feed rates was not examined with the reactor containing a mixture of EDC and β -tri. If one allows for the fact that the rate of decomposition of β -tri into dichloroethylene only depends on the temperature and the ferric chloride content, it is better to work with maximum feed rates for the equipment so as to obtain the smallest possible ratio of β -tri dehydrochlorinated to EDC made.

The upper limit of the feed rates could not be determined in the laboratory apparatus. The difficulties encountered were essentially mechanical; entrainment of liquid by the gases, flooding of the distillation column, entrainment of the Raschig rings Even in the absence of these difficulties it is undoubtedly difficult to determine the maximum rate because if the reaction is not complete in the reactor it can continue in the distillation column where more over, it can form proportionally more β -tri.

EFFECT OF GAS FLOWS

Test No		33	29	31	54
Flows moles/hr	Cl ₂	6	10	20	30
	C ₂ H ₄	6.18	10.1	20.6	30.9
	air	0.12	0.2	0.4	0.6
Average Reactor Temperature °C		82	81	84	79
Degree of Substitution		0.5	0.4	0.6	0.2
C ₂ H ₄ in the vent gas (± 5), %		48-63	25-55	35-40	42
Composition of Reactor Contents g/kg	Fe	0.7	0.8	1.2	1.6
	EDC	941	967	937	889
	β -tri	51	26	55	74
	Other chlorinated compounds	Balance			

M THE EFFECT OF PACKING IN THE REACTOR

The tests were carried out almost completely in a reactor packed with mild steel Raschig rings.

In order to determine if these rings also played the mechanical role in the reaction eg in preventing the formation of large gas bubbles, several tests were carried out in an empty reactor (the reactor contents were artificially dosed with ferric chloride which otherwise was made in situ by corrosion of the metal rings.)

In the absence of the metal rings a slight increase in the degree of substitution was observed. This was not suppressed by placing (in the reactor) metal plates serving to ensure the manufacture in situ of ferric chloride. On the other hand when ethylene and chlorine were mixed before introducing them in the reactor (tests 53 and 54) of the table on page 40)) a slight improvement was observed. (In tests 31 - 52, 53 the chlorine and the ethylene were introduced separately to the reactor). This hypothesis confirmed by the results of a short test carried out in a apparatus without an effective distillation column and the metal rings also being replaced by glass ones. In these conditions the degree of substitution remained small inspite of separate introduction of chlorine and ethylene into the reactor. Apart from this effect of the mixing of the reactants the metal packing favours the contact between the gas and the liquid phase. In addition as already mentioned above it helps to maintain a high ferric chloride content in the circulation. (See table, page 40).

THE EFFECT OF PACKING

Test No		31	52	53		54	22
Flows Moles/hr	Cl ₂	20	20	20	20	30	6
	C ₂ H ₄	20.6	20.6	20.6	20.6	30.9	6.04
	air	0.4	0.4	0.4	0.4	0.6	0.12
Reactant Feeds		Separate			Mixed		Separate
Nature of Packing		Mild Steel Raschig Rings	No Packing	2 Mild Steel Plates 1.65.0 x 20 x 0.5mm			Glass Raschig Rings 10x10 mm
Average Reactor Temperature °C		84	81	84	84	79	84
Degree of Substitution %		0.6	1	0.8	0.4	0.2	0.5
C ₂ H ₄ in Vent Gas (±5)%		35-40	22-35	-	40	40-42	40
Composition of Reactor Contents	Fe	1.2	2.1	1.5	2.1	1.6	0.2
	EDC	937	865	888	859	889	(999)
	β-tri	55	91	81	95	74	(0)
	Other chlorine compounds	Balance					

N THE EFFECT OF BY-PRODUCTS CONTAINED IN THE EDC RECYCLE FROM
CRACKING AND IN THE CRUDE EDC FROM OXYCHLORINATION

Advantages can be taken of the heat released by the chlorination of ethylene to purify the crude EDC from oxychlorination and the EDC recycle from cracking. However, these two streams contain impurities which react in the chlorinator.

The impurities contained in the recycled EDC are numerous and some of them have not yet been identified. Rather than constitute a synthetic mixture we used an effluent from the base of the heavies still (E.182) of the Tavaux vinyl chloride plant (this stream corresponds closely enough to the composition of the impurities which would be fed to the chlorinator). The introduction of these impurities into circulation of the boiling reactor is accompanied by the appreciable evolution of hydrogen chloride which leads to an increase in the apparent degree of substitution.

The liberation of hydrogen chloride is attributed to the dehydrochlorination of chlorine compounds catalysed by ferric chloride. The olefines thus could subsequently undergo a succession of chlorination and dehydrochlorination reactions. As is shown the results given in Table 6 in the appendix, the liberation of hydrogen chloride is especially large when the impurities are introduced and subsequently falls rapidly.

However, several hours are necessary before the apparent degree of substitution returns to normal (tests 59 and 60 of Table 6). The chlorinated substances responsible for these reactions are mainly C₄ compounds such as the chlorobutanes, chlorobutenes, chlorobutadienes present in the recycle of EDC. In agreement with our observations, ICI ANZ established that the residual chlorine present in direct chlorination EDC is eliminated when this EDC is mixed with EDC recycled from the cracking section containing the unsaturated impurities mentioned above (5).

In the case of tests 59 and 60 the reactor contents were almost wholly EDC and β -tri. For these tests the liquid taken off from the base of the heavies still at Tavaux was introduced all at once at the start of the tests. The quantity of liquid introduced represented approximately 5% w/w of the reactor contents in test 59 and 0.5% w/w in test 60. In these two cases 20 hours and 5 hours respectively were necessary for the apparent degree of substitution to return to a value slightly greater than normal.

For test 58 of Table 6, the reactor contents were completely replaced by the liquid taken from the base of the heavies still, the composition of which is given on page 43. The degree of substitution in this test was high (8-15%), and it increased still more (32%) when mild steel Rashig rings were introduced into the reactor. It is probable that this resulted in an increase in the ferric chloride content which accelerated the dehydrochlorination reactions.

In practice the impurities would be introduced continuously into the reactor and a continuous purge from the reactor circulation should be considered. In trial 61-1 about 20 ccs per hour of liquid from the base of the heavies still were introduced continuously over a period of 31 hours. This flow represented 1.2% w/w of the EDC manufactured. Under these conditions the degree of substitution steadied out at between 2.5 and 2.7%. After the test had gone on for 31 hours the introduction of the impurities was stopped; only eleven hours later did the degree of substitution return to its normal value - 0.4% (test 61-2).

A much larger quantity of impurities (approximately 2.5% w/w of the amount of EDC made in the chlorinator) was introduced during test 61-3; simultaneously a continuous purge was taken from the reactor circulation (300 cc per hour for 14 hours and 200 ccs/hour for 23 hours or 19% and 12.5% respectively of the EDC produced. Under these conditions the degree of substitution stabilized out at about 3.2 to 3.5% during the 37 hours of operation; towards the end however the degree of substitution had a slight tendency to climb. It should be noted that if the purge from the chlorinator represented about 20% of the EDC manufactured by chlorination in a balanced crack (?-balance between direct and oxychlorination) and for a depth of crack of 50% this purge represented about 5% of the total EDC treated in the boiling reactor.

The effect of the impurities on the chlorination reaction itself is very difficult to establish under our conditions. However it appears that the chlorine and the ethylene still react together completely; nevertheless a slightly higher quantity of ethylene is found in the vent gas than under normal conditions. The presence of this excess of ethylene is connected with the consumption of part of the chlorine by reaction with the impurities.

When the vent gas flow is appreciably increased by the liberation of substantial volumes of HCl, a small amount of chlorine is carried over to the scrubber. At first sight the real degree of substitution does not seem to have increased.

In the special case of introducing crude oxychlorination EDC on its own, which contains fewer impurities than EDC recycled from cracking, into the boiling reactor, an appreciable increase in the HCl content of the vent gases as compared with that observed in the case of chlorination without the introduction of impurities could not be established.

COMPOSITION OF THE SAMPLE TAKEN FROM THE BASE OF THE HEAVIES STILL AT
TAVAU (By Chromatography)

Constituents	g/kg	Constituents	g/kg
chloroform	0.2	trans 1,4-dichloro-2-butene	6.4
unknown	0.2	1,1,2,2-tetrachloroethane	47
unknown (I 1120)	0.1	pentachloroethane	12
1,2-dichloroethane	129	unknown	35
1,1,1-trichloroethane	0.1	hexachloroethane	15
benzene	0.1	1,1,2,4,4-pentachlorobutadiene)	13
trichloroethylene	0.1	unknown)	
1-bromo-2-chloroethane	7.6	hexachlorobutadiene	1.7
unknown	0.8	1,2,3-trichlorobenzene	5.3
unknown	8.1	unknown	0.2
1,1-dichlorobutane	0.2	unknown	0.3
1,1,2-trichloroethane	418	unknown (I 1094)	2.4
1,2-dichloro-3-butene)	2.8	1,2,3,4-tetrachlorobenzene	0.7
2,3-dichlorobutane)		pentachlorobutyne	0.7
unknown	19	unknown (I 1095)	0.1
perchloroethylene)	73	pentachlorobenzene	0.2
unknown)		unknown (I 1097)	0.6
unknown	0.2	unknown (I 1102)	0.6
1,2-dichlorobutane	3.1	unknown	2.4
cis 1,3-dichloro-2-butene)	87	unknown	0.8
1,3-dichloro-1-butene)		unknown	1.9
1,3-dichlorobutane)		hexachlorohexadiene-yne	0.3
unknown	6.7	unknown (I 1107)	2.3
trans 1,3-dichloro-2-butene)	80	unknown	0.9
(cis or trans)1,1-dichloro-2-butene)			
unknown	0.6		
1,1,2,2-tetrachloroethane	5.7		
monochlorobenzene	4.9		
cis 1,4-dichloro-2-butene	0.5		

Heavy Residue (by evaporation to dryness), g/kg: ~ 80

O EQUIPMENT CORROSION

Metal test pieces 50 mm x 10 mm x 1 mm were placed in several positions of the laboratory apparatus. These are shown in Fig 3.

The test pieces were not removed from the apparatus while it was off line. Consequently the rate of corrosion calculated from the effective lengths of the run includes the corrosion of the test pieces during shut downs. In fact the rates of corrosion have mainly a relative value (see Table below). Among the materials tested tantalum showed the best resistance to corrosion followed in order by inconel, nickel and Uranus SD stainless steel. The performance of Martin and Thomas mild steels was clearly less than that of the materials mentioned above.

Two metal tubes, one in mild steel and the other in inconel were placed in the rectifying column condenser (vertical orientation). Very marked local corrosion of the two tubes was observed at the position of the interface (of separation) between the condensing vapours and the residual gases (see Fig 7). This very severe corrosion is probably due to the condensation of the EDC - water azeotrope at the level of the interface with the formation of a thin milky film at the position of the interface.

RATE OF CORROSION IN VARIOUS PLACES IN THE APPARATUS

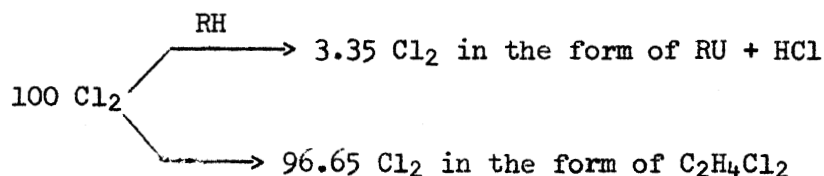
Place and Number of Sample Pieces (See Fig 2)		Reactor Liquid Phase 1	Reactor Gas Phase 2	Top of the Heavies Still 3	Gas at the Exit from the apparatus before the scrubber 4
Effective length of Test (hrs)		209			185
Length test pieces remained in Apparatus (days)		63			57
Loss in Weight g/m ² day	Mild Steel (Martin)	13.7	10.5	3.14	17.3
	Mild Steel (Thomas)	13.4	11.6	2.50	14.0
	URANUS SD (Stainless steel)	1.94	0.14	0.08	2.89
	Nickel	0.75	0.22	0.42	0.11
	Inconel	1.02	0.29	0.14	0
	Tantalum	0.04	0.09	0.02	0.05

P CONSUMPTION OF CHLORINE BY REACTION WITH IMPURITIES IN THE EDC RECYCLED FROM CRACKING AND IN THE CRUDE EDC FROM OXYCHLORINATION

The by-products contained in the EDC recycled react with chlorine when they are introduced into the reactor. The chlorine thus consumed can be estimated from the quantity of HCl liberated when these by-products are introduced; this is a rough guess because these chlorination reactions are not necessarily substitutions (ie could be additions). The fraction of the chlorine consumed, calculated on this basis probably represents a maximum value: firstly without the intervention of chlorine, only by the action of ferric chloride (some of) the impurities give off large amounts of HCl, and secondly the HCl produced by the conversion of EDC into β -tri is not subtracted from the total HCl. Furthermore the higher the number of chlorine atoms already present in the molecule of the unsaturated compound the slower the rate of chlorine addition.

For an apparent degree of substitution of 3.35% (average of 3.2 to 3.5%) the loss in chlorine is calculated as follows (See III-N):

- 1 Loss expressed as kg Cl₂/kg EDC manufactured by direct chlorination:



from where

$$\frac{0.0335}{0.9665} \times \frac{71}{99} \approx 0.025 \text{ kg Cl}_2 \text{ per kg of EDC manufactured}$$

- 2 Loss expressed as kg Cl /kg of total EDC manufactured:

For a balanced process half the EDC is manufactured by oxy-chlorination and one has a 99% yield based on HCl for oxy-chlorination:

$$0.025 \times \frac{1}{1.99} \approx 0.0125 \text{ kg Cl}_2 \text{ per kg of total EDC manufactured}$$

- 3 Loss expressed as kg Cl /kg VC manufactured (for a yield of 100%):

$$0.0125 \times \frac{99}{62.5} = 0.0198 \text{ kg Cl}_2 \text{ per kg of VC manufactured}$$

4 Loss of Chlorine Expressed as Belgian Francs/kg VC:

$$0.0198 \times 2.75 = 0.055 \text{ Belgian Francs per kg of VC}$$

This loss is an upper limit; it must be compared with the gain resulting in the saving of steam.

For $C = 0.5$ one can recover 25.1 kg cal/mol of VC manufactured (diagram 2-A) from where the saving based on steam at 120 Belgian Francs per te:

$$\frac{25.1 \times 16.0}{530} \times 0.120 = 0.091 \text{ Belgian Francs per kg of VC}$$

The saving is therefore $0.091 - 0.055 = 0.036$ Belgian Francs per kg of VC if the EDC recycled from cracking is returned to the chlorinator.

When only oxychlorination EDC (at 83°C) is returned to the chlorinator the savings which can be achieved depend on the reflux ratio. If as at Tavaux $R = 0.77$ the saving is

$$\frac{0.091 \times 7.8 \times 1.77}{25.1} = 0.05 \text{ Belgian Francs/kg of VC}$$

One can thus recover 55% of the heat produced in the chlorination reaction. (The additional heat makes it necessary to increase the reflux ratio to 2.18).

In the latter case, however, an allowance must be made for a distillation column to eliminate the heavies in the EDC recycled from cracking.

Q PURGE FROM THE REACTOR AND RECOVERY OF 1,1,2-TRICHLOROETHANE

The use of the direct chlorinator as a boiler for the heavies still leads to an accumulation of heavies in the reactor. C_2 , C_4 and C_6 chlorinated compounds undergo a series of dehydrochlorination - chlorination reactions which give rise to substances which are only slightly soluble. In addition carbon and ferric chloride build up. Furthermore the β -tri which could be recovered for the manufacture of vinylidene chloride is slowly converted into 1,1,2,2-tetrachloroethane which is not recovered and is eliminated with the tars (see II-I).

It is therefore essential to maintain a continuous purge from the reactor. It is therefore necessary to allow for equipment for recovering the EDC and the β -tri associated with EDC purification. A unit for reaction, purification and recovery is proposed in Fig 8. The purge to be treated must be limited (to a maximum) so as not to penalize excessively chlorination in a boiling reactor.

At Tavaux heavies production currently represents about 1% of the (quantity of) EDC purified in the heavies still. If one assumes a reactor circulation containing 20% heavies (β -tri etc), the purge

required represents $\frac{1}{20}$ ie 5% of the total EDC leaving the reactor. Given that the recovered EDC does not have to be completely free from β -tri the amount of steam consumed by the recovery remains negligible compared with the steam saved by the boiling reactor.

It was not possible to determine the optimum purge in the laboratory. The behaviour of the reaction remained excellent when the liquid in the reactor circulation contained about 9% w/w of β -tri, 7% heavies and 84% EDC.

This liquid containing a lot of iron was distilled at atmospheric pressure. Starting with 3.3 ltrs of purged liquid, 2.7 ltrs were recovered before the residue went solid. Recovery of EDC and β -tri in the purge liquid by distillation can therefore be ruled out. It is therefore necessary to recover them by steam distillation as currently carried out at Tavaux. Due to the large amount of ferric chloride in the purge liquid, it will perhaps be necessary to carry out a preliminary acid wash.

R PERFORMANCE OF THE ETHYLENE AND CHLORINE ANALYSERS

For information we consider it to be worthwhile to give some indications on the working of the analysers which were used to measure continuously the chlorine and ethylene contents of the vent gases. These must be conditioned so as to eliminate the entrainment of droplets and to avoid possible condensation in the sample systems.

1 Ethylene Analyser

Ethylene content was determined by a non dispersive infra-red analyser (made by 'Controle de Chauffe - ONERA type 8) under licence). This analyser worked satisfactorily during the tests. However calibration was required from time to time. The newer instruments of this type (Maitrak type UNOR - ONERA licence) tried at Tavaux are of even better quality.

2 Chlorine Analyser

Chlorine content of the vent gas was measured with a UV photometer with an interferential filter made at NOH. In practice the chlorine content of the vent gas could be followed by measurement of the free chlorine in the water from the scrubber. The use of the analyser however made it possible to detect sudden changes in the course of the reaction. Apart from frequent fouling of the sample cell due mainly to condensation, the following difficulties were encountered:

- instability of the recorder trace (background noise) which was remedied by installing a voltage stabilizer and increasing the signal at the phototubes.

- zero drift at constant sensitivity which necessitated frequent adjustment (new instruments are equipped with automatic zeroing).

S CONCLUSIONS

The chlorination of ethylene in a liquid medium at its boiling point can be carried out with yields comparable to those obtained in the conventional process. The chlorinator can be used as a reboiler for a distillation column and the EDC thus obtained is free from ferric chloride and 1,1,2-trichloroethane.

The introduction of crude oxychlorination EDC and of the EDC recycled from the cracking section of the plant into the chlorinator makes it possible to recover the heat of reaction. However the impurities present in the EDC recycled from cracking react partially with the chlorine. The chlorine yield is thus lowered by about 3% which from an economic point of view corresponds to about 60% of the heat which can be recovered in a balanced oxychlorination process.

In principle the introduction of crude oxychlorination EDC on its own into the chlorinator should not lower the chlorine yield. In this case one can recover 55% of the heat liberated by the reaction with a reflux ratio of 0.77.

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MASS BALANCE DIRECT CHLORINATOR AT TAVAU (1) (Taveaux Bulletin CL/CBa of 26.2.69)

Table 1

Material Analysed	Ethylene		Chlorine		EDC Product		Vent Gas		Organics Collected	
	Weight %	Flow kg/h	Weight %	Flow %	ppm Weight	Flow kg/h	% Weight	Flow kg/h	ppm Weight	Flow kg/h
Components										
Ethylene	99.95	6.261					7.20	104.7		
Ethane	0.05	3					0.21	3		
Inerts			1.51	241			16.58	241		
Hydrogen Chloride					410(2)	7.1	1.01	14.7		
Chlorine			98.49	15.716	184(2)	3.2	0.41	5.9		
Vinyl Chloride							0.01	0.22	10	0.22
Ethyl Chloride					398	8.2	0.19	2.80	503	11.00
1,1-Dichloroethane					50	1.0		0.11	50	1.11
Chloroform					65	1.3		0.11	65	1.41
1,2-Dichloroethane					99.64(3)	20,692.3	74.29	1.080	99.64(3)	21,772.3
1-Bromo-2-chloroethane					130	2.7			123	2.70
1,1,2-Trichloroethane					2870	59.6	0.08	1.10	2,777	60.70
1,2-Dichlorobutane					120	2.5			119	2.50
Total		6.264		15.957		20,767.6 (4)		1,453.6		21,851.9

- (1) Based on flows measured and analyses carried out over a period of 3 days
Consumption of C₂H₄ per kg of organics : 0.287 (≈ 98.5-99%)
Cl₂ " " : 0.720 (≈ 99.5-100%)
(2) in gm/m³ (3) weight % (4) Excluding HCl & Cl₂

TABLE 4

THE DEVELOPMENT OF THE DEGREE OF SUBSTITUTION AND THE COMPOSITION OF THE REACTOR CIRCULATION AS A FUNCTION OF TIME
FOR A CIRCULATION CONSISTING INITIALLY OF PURE EDC

Time from Start (hrs)	1.5	2.5	5	8	9	12	15	16	18	21	22	23	25
Degree of Substitution %	0.8	0.5	0.3	0.1	0.3	0.3	<0.05	<0.05	-	0.2	0.2	0.3	0.2
% C ₂ H ₄ in Vent Gas	62	14	34	-	-	-	-	-	-	-	-	-	-
Composition of the Circulation g/kg (EDC made $\approx$ 50 kg)													
Time from Start	Fe	EDC	C ₂ H ₆ Cl	CH ₂ BrCH ₂ Cl	β Tri	Sym Tetra	C ₂ HCl ₅	C ₂ Cl ₈	C ₂ Cl ₁₄	Chloro-benzenes			
0 hours	-	$\approx$ 1000	-	-	-	-	-	-	-	(Benzene)			
18 hours	0.33	Balance	0.2	1.0	6.2	1.2	1.7	0.1	0.1	3.7			
25 hours	0.61	Balance	<0.05	1.3	29.4	3.8	1.7	0.4	2.8	2.5			

OPERATING CONDITIONS

Flow, Moles/hr : C₂H₄ 20.6
Cl₂ 20
Air 0.4

Reactor Temperature °C : 81
Type of Reactor Packing : Mild Steel Raschig rings
Test Programme : Generally the tests lasted 8 hours with breaks of 16 hours

TABLE 5

EFFECT OF β -Tri CONTENTS OF REACTOR CONTENT ON THE DEGREE OF SUBSTITUTION

Test No		46			47			48	49	57			
Flows Moles/hr	Cl ₂	20	3	20	0	1	0	0	10	(30)	20	20	20
	C ₂ H ₄	20.6	3.09	20.6	0	0	0	0	10.3	30.9	20.6	20.6	20.6
	Air	0.4	0.06	0.4	0.7	0.7	2	4	0.2	0.6	0.4	0.4	0.4
Average Reactor Temp °C		96	92	94	91	91	90	89	94	100	93	93	92
Apparent Degree of Substitution %		2.3	3.6	2.3	(0.2)	(3)	(2.2)	(0.9)	3.5	5	0.9	0.9	0.8
C ₂ H ₄ in the Vent Gas (±5)%		45	65	20	-	-	-	-	63	55	22	16	18
Composition of the Reactor Contents	Fe	0.15 to 0.37			-	-	-	0.6	-	1.1	0.4	0.5	0.8
	EDC	328 to 323			-	-	-	260	-	169	439	468	422
	β-Tri	637 to 626			-	-	-	663	-	565	427	411	435
	Sym Tetrachloroethane	21 to 39			-	-	-	56	-	236	120	108	127
Penta, Hexa, Tri, Per etc		Balance											
Cis 1,2 Dichloroethylene collected in the gases g/hr		-	-	0.3	-	-	-	40	-	15	0.6	0.6	0.6

TABLE 6
EFFECT OF BY-PRODUCTS CONTAINED IN THE EDC RECYCLED FROM
CRACKING AND THE CRUDE EDC FROM OXYCHLORINATION

Test No	59				60				61-1	61-2	61-3	58				
Flows Moles/hr	20				20				20				20			
	20.6				20.6				20.6				20.4			
	0.4				0.4				0.4				0.4			
Length of Test (hours)	1	10.5	19	0	0.5	1	3.5	4.5	see text				11.5	20	32	45
Average Reactor Temp °C	84	84	84	81	81	81	81	81	83	84	83	112	112	111	117	
Degree of Substitution %	12	1.4	0.8	12	2.2	1.2	1.2	0.9	2.6	0.4	3.3	8	10	15	32	
C ₂ H ₄ in the Vent ($\pm$ 5) %	-	-	-	-	-	-	-	-	50	50	50	73	-	60	76	
Fe	-	-	-	-	-	-	-	-	4.3	4.1	4	0.5	-	-	5.8	
Composition of Reactor Contents g/kg	$\approx$ 750	770	810	-	-	-	-	-	$\approx$ 850	853	863	92	$\approx$ 100	$\approx$ 100	40	
β Tri	$\approx$ 150	144	108	-	-	-	-	-	$\approx$ 100	84	85	273	$\approx$ 300	$\approx$ 300	246	
Others	Balance			-	-	-	-	-	$\approx$ 50	Balance	Balance	Balance				
Chlorine in the Vent Gas Moles/hour	0.01	0	0	-	-	-	-	-	-	-	-	0.02	0.37	0.22	Traces	